Date-22/07/2025

Practical No. 2

Aim: Create a Windows Virtual Machine using Microsoft Azure.

Theory:

What is a Virtual Machine?

A Virtual Machine (VM) is a software-based simulation of a physical computer. It runs an operating system and applications just like a real computer, but instead of using physical hardware directly, it uses virtualization software (called a hypervisor) to share resources of a host machine.

In simple terms:
A VM = A computer inside another computer.

How does a virtual machine work?

A Virtual Machine (VM) runs on top of a hypervisor, which is software that divides the host computer's hardware (CPU, RAM, storage, network) into virtual parts. Each VM gets its own share of these resources and runs its own operating system (called the guest OS) as if it were a real, separate computer.

The hypervisor ensures:

- Resource allocation (decides how much CPU, memory, etc. each VM gets).
- Isolation (if one VM crashes, others stay safe).
- Translation (guest OS instructions are mapped to real hardware operations).

Because VMs are stored as files, they can be paused, cloned, or moved easily between machines or cloud environments.

What benefits do VMs offer?

A. **Cost Efficiency:** Multiple VMs can run on a single physical server, reducing hardware needs.

B. **Flexibility:** You can run different OS and software on the same host.

C. **Isolation:** Problems in one VM don't affect others or the host.

D. **Scalability:** Resources (CPU, RAM, storage) can be increased or decreased easily.

E. **Testing & Development:** Safe sandbox environments for trying software without risks.



F. **Backup & Recovery:** Easy to clone, move, or restore VMs if something goes wrong.

G. **Security:** Each VM is isolated, reducing risks of cross-contamination.

H. **Portability:** VM images can be transferred across systems or cloud environments.

What are future VM trends?

1. **Cloud-first adoption**: More organizations are moving VMs to cloud platforms like Azure, AWS, and Google Cloud.
2. **Containers integration**: Containers (e.g., Docker, Kubernetes) are becoming popular, and future trends show a mix of VMs and containers for maximum flexibility.
3. **Automation & AI**: Smart management of VMs with AI to optimize resource usage.
4. **Serverless and edge computing**: Some workloads will shift from VMs to serverless and edge technologies, but VMs will still remain important.
5. **Improved security**: Future VMs will have stronger isolation and protection against cyber threats.

Steps of creatin VM using Microsoft Azure.

Steps to create a VM in **Azure portal**:

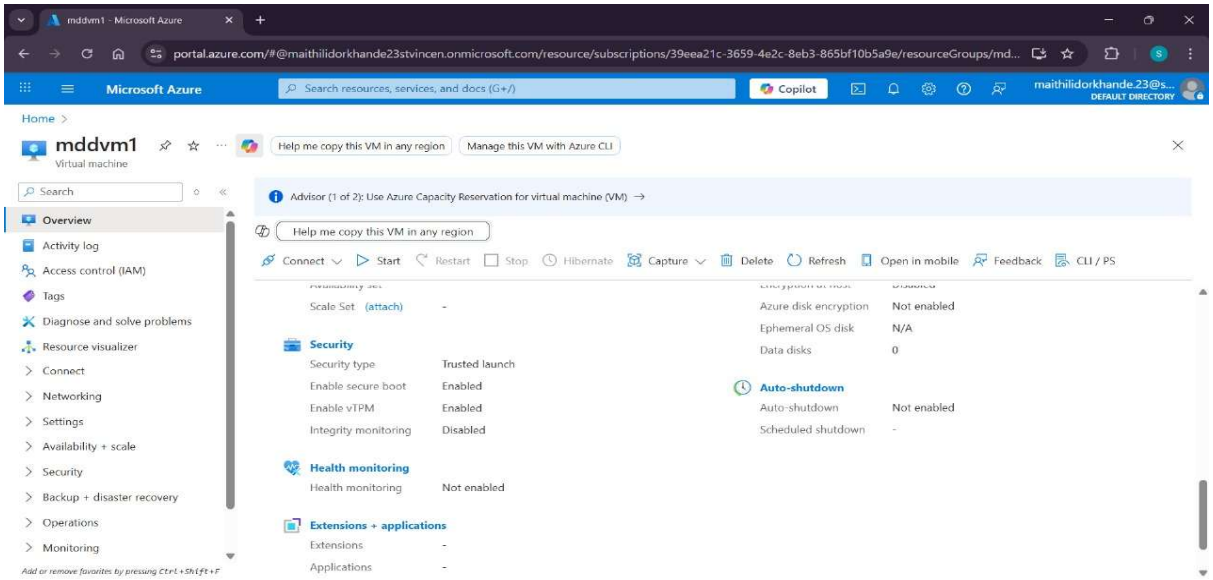
1. **Login to Azure Portal**
   - Go to portal.azure.com and sign in with your account.
2. **Create a Resource**
   - Click on **"Create a resource"** from the dashboard.
3. **Choose Virtual Machine**
   - From the available services, select **"Virtual Machine"** under **Compute**.
4. **Fill in Basic Details**
   - Subscription: Select your Azure subscription.
   - Resource Group: Create a new one or select an existing group.
   - VM Name: Give a unique name.
   - Region: Choose the data center region.
   - Image: Select OS (e.g., Windows Server, Ubuntu).
   - Size: Choose the VM size (CPU, RAM).

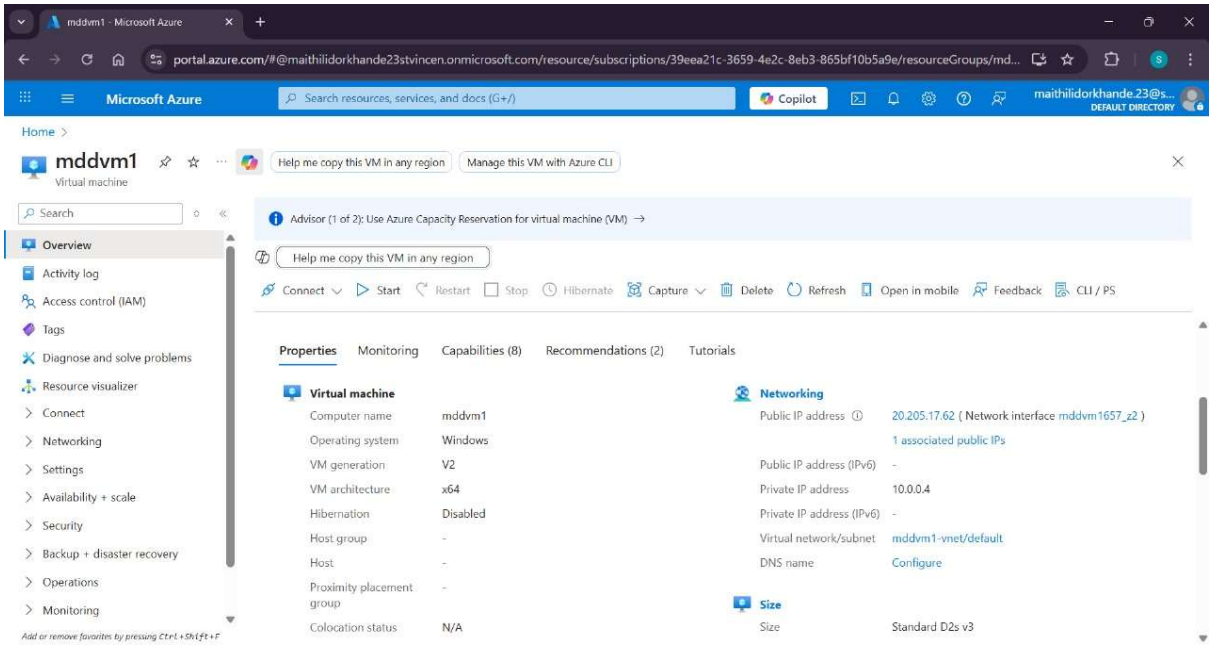


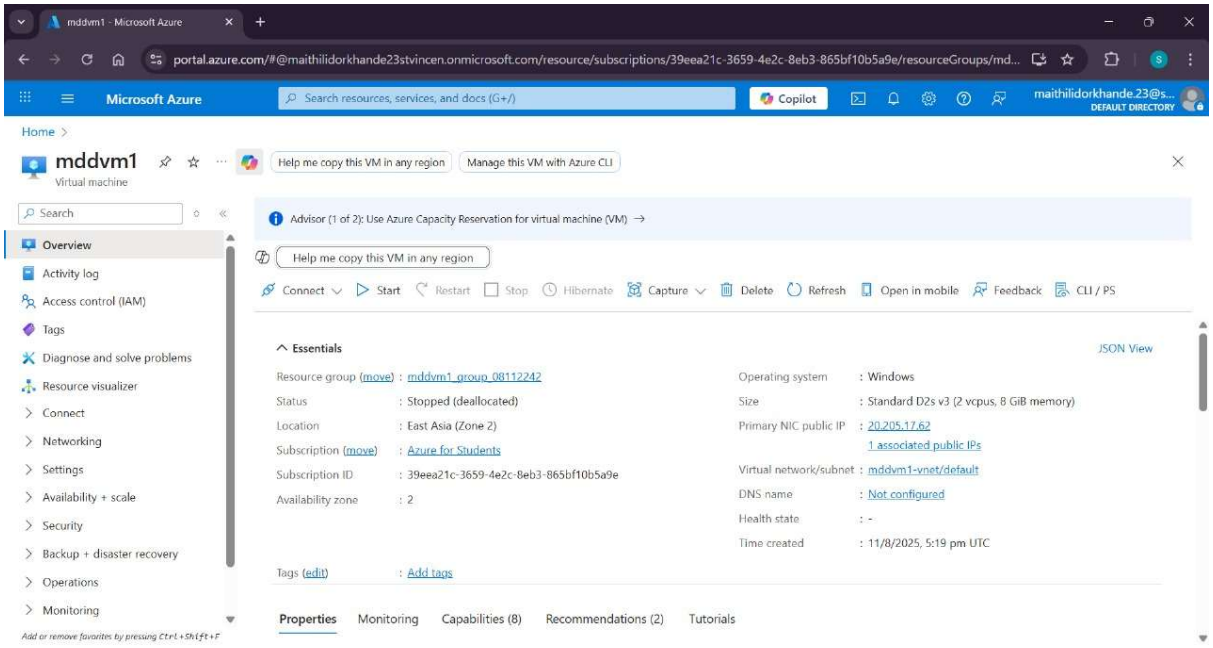
- Authentication: Select password or SSH key for login.

5. **Configure Disks & Networking**
   - Choose disk type (Standard HDD, SSD).
   - Set up network settings (Virtual Network, Subnet, Public IP).
6. **Review + Create**
   - Review all settings.
   - Click **"Create"** to deploy the VM.
7. **Access the VM**
   - Once created, connect to the VM using **RDP (Windows)** or **SSH (Linux)** with the provided IP address.

Screenshots (SS):




mddvm1 - Microsoft Azure
portal.azure.com/#@maithilidorkhande23stvincen.onmicrosoft.com/resource/subscriptions/39eea21c-3659-4e2c-8eb3-865bf10b5a9e/resourceGroups/md...
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
maithilidorkhande.23@s...
DEFAULT DIRECTORY
Home
mddvm1
Virtual machine
Help me copy this VM in any region
Manage this VM with Azure CLI
Search
Advisor (1 of 2): Use Azure Capacity Reservation for virtual machine (VM)
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Connect
Networking
Settings
Availability + scale
Security
Backup + disaster recovery
Operations
Monitoring
Add or remove favorites by pressing Ctrl+Shift+F
Help me copy this VM in any region
Connect
Start
Restart
Stop
Hibernate
Capture
Delete
Refresh
Open in mobile
Feedback
CLI / PS
Properties
Monitoring
Capabilities (8)
Recommendations (2)
Tutorials
Virtual machine
Computer name mddvm1
Operating system Windows
VM generation V2
VM architecture x64
Hibernation Disabled
Host group -
Host -
Proximity placement group -
Colocation status N/A
Networking
Public IP address 20.205.17.62 (Network interface mddvm1657_z2)
1 associated public IPs
Public IP address (IPv6) -
Private IP address 10.0.0.4
Private IP address (IPv6) -
Virtual network/subnet mddvm1-vnet/default
DNS name Configure
Size
Size Standard D2s v3


mddvm1 - Microsoft Azure
portal.azure.com/#@maithilidorkhande23stvincen.onmicrosoft.com/resource/subscriptions/39eea21c-3659-4e2c-8eb3-865bf10b5a9e/resourceGroups/md...
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
maithilidorkhande.23@s...
DEFAULT DIRECTORY
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mddvm1
Virtual machine
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Help me copy this VM in any region
Connect
Start
Restart
Stop
Hibernate
Capture
Delete
Refresh
Open in mobile
Feedback
CLI / PS
Essentials
JSON View
Resource group (move) : mddvm1_group_08112242
Status : Stopped (deallocated)
Location : East Asia (Zone 2)
Subscription (move) : Azure for Students
Subscription ID : 39eea21c-3659-4e2c-8eb3-865bf10b5a9e
Availability zone : 2
Operating system : Windows
Size : Standard D2s v3 (2 vcpus, 8 GiB memory)
Primary NIC public IP : 20.205.17.62
1 associated public IPs
Virtual network/subnet : mddvm1-vnet/default
DNS name : Not configured
Health state : -
Time created : 11/8/2025, 5:19 pm UTC
Tags (edit) : Add tags
Properties
Monitoring
Capabilities (8)
Recommendations (2)
Tutorials

Conclusion:

A Windows Virtual Machine named mddvm1 was successfully created in Microsoft Azure, providing a secure, scalable, and flexible environment for running applications and managing resources in the cloud.

Attendance	Performance	Viva	Total
/3	/7	/5	/15

